



2007

**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Biology

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- Write your Centre Number and Student Number at the top of pages 9, 13, 17 and 19

Total marks – 100

Section I Pages 2–20

75 marks

This section has two parts, Part A and Part B

Part A – 15 marks

- Attempt Questions 1–15
- Allow about 30 minutes for this part

Part B – 60 marks

- Attempt Questions 16–27
- Allow about 1 hour and 45 minutes for this part

Section II Pages 21–31

25 marks

- Attempt ONE question from Questions 28–32
- Allow about 45 minutes for this section

Section I
75 marks

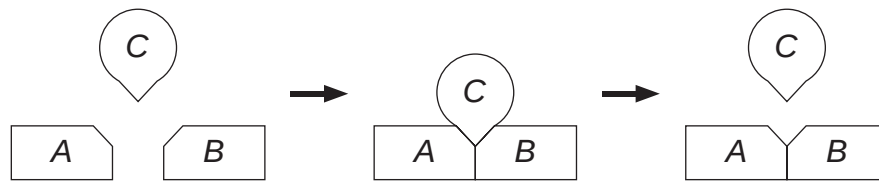
Part A – 15 marks

Attempt Questions 1–15

Allow about 30 minutes for this part

Use the multiple-choice answer sheet for Questions 1–15.

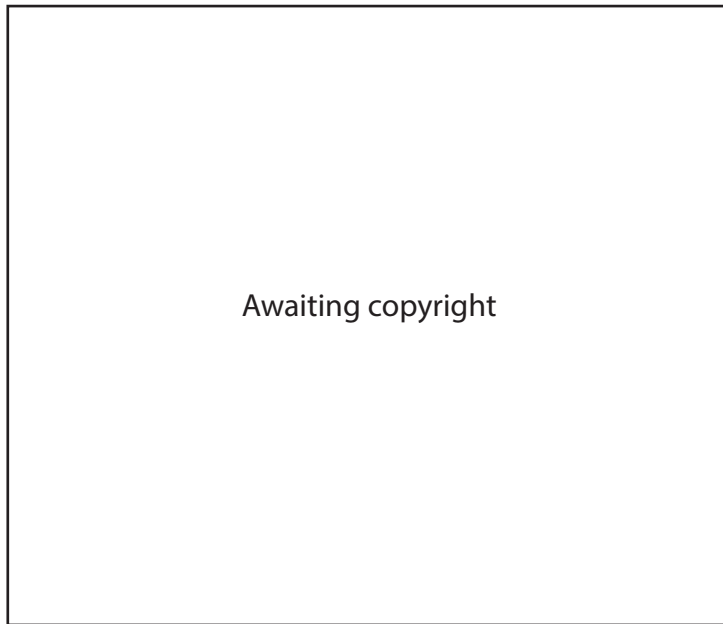
- 1** The diagram shows one example of enzyme action as demonstrated by the ‘Lock and Key’ model.



Which part of the diagram represents the substrate?

- (A)
- (B)
- (C)
- (D)
- 2** Why do cells contain many different enzymes?
- (A) Enzymes are temperature specific.
- (B) Enzymes are specific in their action.
- (C) Enzymes are sensitive to pH changes.
- (D) Enzymes are sensitive to substrate concentration.

- 3 This is a longitudinal section of plant stem ($\times 200$).



What is the name and function of the tissue labelled W?

	<i>Name</i>	<i>Function</i>
(A)	Xylem	Transports water and mineral ions
(B)	Phloem	Transports water and mineral ions
(C)	Xylem	Transports simple sugars
(D)	Phloem	Transports simple sugars

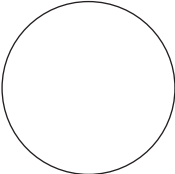
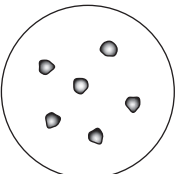
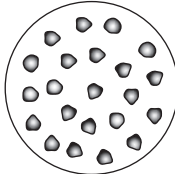
- 4 The Wollemi Pine (*Wollemia nobilis*) is easily killed by the fungus *Phytophthora* which lives in the soil. The last small population of Wollemi Pines grows in a remote part of a national park in NSW. Scientists studying this natural population use strategies to prevent the trees becoming infected with *Phytophthora*.

Which procedure would be most effective in preventing the spread of this fungus to the Wollemi Pines?

- (A) Inspecting soil samples in the area
- (B) Commercially producing and distributing the Wollemi Pine
- (C) Washing soil from scientists' shoes before they walk in the area
- (D) Preventing the importation of infected Wollemi Pines into Australia

- 5 Students performed an investigation to compare the effectiveness of two water treatments for purifying pond water.

Three samples of pond water, *A*, *B* and *C*, were collected and each used to inoculate an agar plate. The plates were incubated at 25°C and examined three days later. The number of visible bacterial colonies on each plate was counted and the results tabulated.

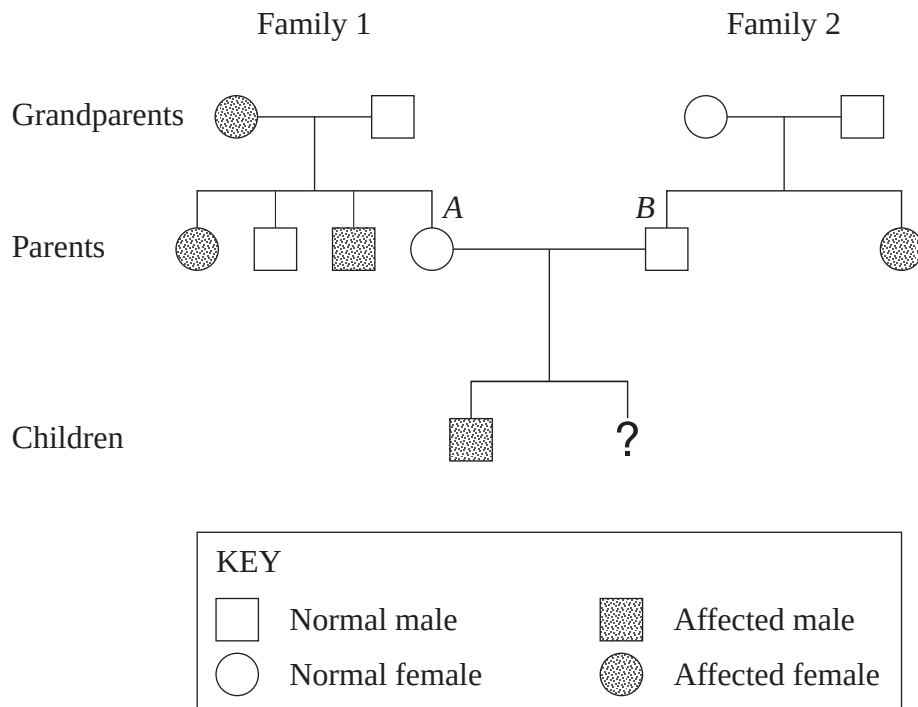
<i>Sample</i>	<i>A</i>	<i>B</i>	<i>C</i>
<i>Treatment</i>	5 grams of pool chlorine per litre of water	Boiling for one minute	No treatment
<i>Number of visible bacterial colonies</i>	0 	6 	22 

What is the dependent variable in this investigation?

- (A) The use of a control sample
 - (B) The number of visible bacterial colonies
 - (C) The use of sterile agar plates for each sample
 - (D) Treating the water by boiling or adding pool chlorine
- 6 Which biological term is best described by ‘engulfing and destruction of bacteria or other foreign bodies’?
- (A) Vaccination
 - (B) Phagocytosis
 - (C) Antibody production
 - (D) An inflammation response

- 7 Which leaf structures are adaptations to assist in the conservation of water?
- (A) Central vein, irregular leaf shape
 - (B) Large air spaces, pointed leaf tip
 - (C) Spongy mesophyll, vascular bundle
 - (D) Sunken stomates, thick waxy cuticle
- 8 Which observations can be used to demonstrate Koch's contribution to understanding the cause of disease?
- (A) Polio vaccinations trigger an immune response.
 - (B) Some mosquitoes carry a pathogen that is often fatal to people.
 - (C) A lack of vitamin C is found in all people suffering the nutritional disease scurvy.
 - (D) The bacteria, *Helicobacter pylori*, is present in the stomach of all people diagnosed with stomach ulcers.
- 9 Current reproductive techniques can be used to alter the genetic composition of a population. Some of these methods were also used in the nineteenth century by Gregor Mendel.
- How did Mendel use reproductive techniques in his experiments?
- (A) He artificially inseminated the pea plants to achieve wrinkled seeds.
 - (B) He cloned the pea plants with round seeds to increase their food supply.
 - (C) He created a transgenic species by mixing tall pea plants and short pea plants.
 - (D) He artificially pollinated the pea plants to test for different genotypes in the offspring.
- 10 Which statement best describes the relationship between proteins and polypeptides?
- (A) Proteins are composed of polypeptides.
 - (B) Polypeptides are composed of proteins.
 - (C) Proteins, unlike polypeptides, are composed of amino acids.
 - (D) Polypeptides, unlike proteins, are composed of amino acids.

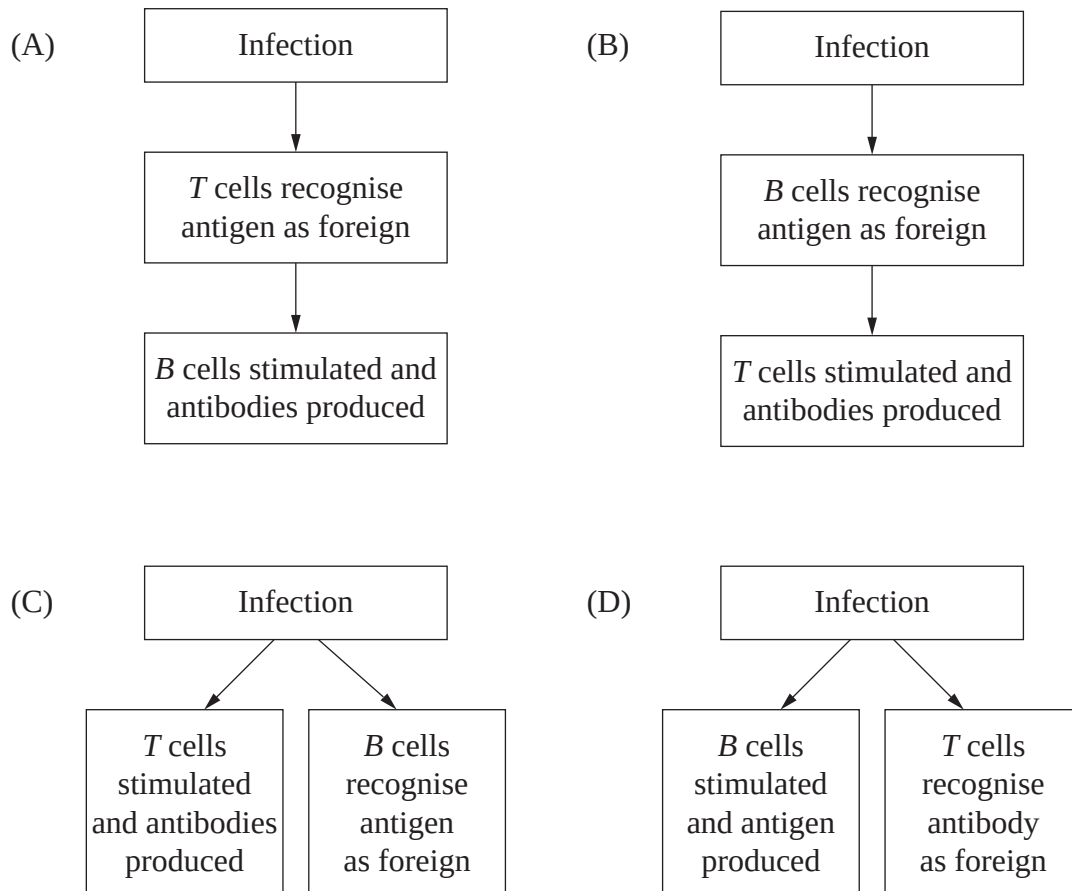
- 11 The family tree shows the inheritance of a genetic characteristic.



What is the probability that a daughter of parents *A* and *B* would be affected?

- (A) 0%
- (B) 25%
- (C) 50%
- (D) 100%

- 12 Which flowchart correctly shows an interaction between *B* and *T* lymphocytes during an immune response?



- 13 The effectiveness of a new insecticide was tested on a large population of mosquitoes over a number of breeding cycles. At first the population of mosquitoes was reduced dramatically by the use of the insecticide. After a number of breeding cycles the population then began to increase until the insecticide appeared to have little effect.

How would the Darwin/Wallace theory of evolution by natural selection explain these observations?

- (A) Some of the original population were isolated from the insecticide as a control group.
- (B) Some of the original population had already reproduced before the insecticide was used.
- (C) Some of the original population were resistant to the insecticide and passed this on to their offspring.
- (D) Some of the original population adapted to the insecticide and survived to produce offspring.

- 14** At the end of a marathon race a runner's body is dehydrated.

How does the body control the two hormones, ADH and aldosterone, to help to re-establish normal water balance?

- (A) ADH is released and aldosterone is inhibited.
 - (B) ADH is inhibited and aldosterone is released.
 - (C) Both ADH and aldosterone are released.
 - (D) Both ADH and aldosterone are inhibited.
- 15** How have Walter Sutton and Theodor Boveri contributed to the understanding of inheritance?
- (A) By determining the structure of DNA
 - (B) By improving knowledge of sex linkage
 - (C) By demonstrating the effect of environment on phenotype
 - (D) By identifying the importance of chromosomes in inheritance

Biology

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Centre Number

Section I (continued)

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Student Number

Part B – 60 marks**Attempt Questions 16–27****Allow about 1 hour and 45 minutes for this part**

Answer the questions in the spaces provided.

Marks**Question 16 (3 marks)**

A student working in a restaurant kitchen is required to wear disposable gloves and hat when preparing food.

- (a) Explain how this practice assists in the control of disease. **2**

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- (b) Identify another hygiene practice that reduces the risk of infection. **1**

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Question 17 (3 marks)

Construct a table to distinguish between the environmental conditions experienced by marine and fresh water fish, and the resulting urine production of each fish. **3**

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Question 18 (4 marks)

(a) Name a blood product extracted from donated blood and outline how it could be used to restore normal body function. **2**

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(b) Propose TWO reasons why research is needed to develop alternatives to donated blood. **2**

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Question 19 (6 marks)

- (a) Name ONE example of a disease caused by a macro-parasite. **1**

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- (b) List TWO features of prions that distinguish them from protozoans. **2**

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- (c) Most pathogens must first be transmitted to and enter the human body before they trigger an immune response. **3**

Relate this statement to a named infectious disease you have studied.

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Centre Number

Section I – Part B (continued)

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Student Number

Question 20 (6 marks)

Please turn over

Question 20 (6 marks)

- (a) The table shows that 20% of the bases in a section of double-stranded DNA are adenine (A). 3

Complete the table below by identifying the other three base types and calculating the percentage of each base type in the section of double-stranded DNA.

<i>Bases</i>	<i>Percentage (%)</i>
A	20

- (b) Construct a simple flowchart to describe the process of DNA replication. 3

Question 21 (6 marks)



- (a) Which disease in the table is the most dangerous? Give TWO reasons for your answer. 2

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- (b) Evaluate the effectiveness of a vaccination program for ONE named disease from the table. 4

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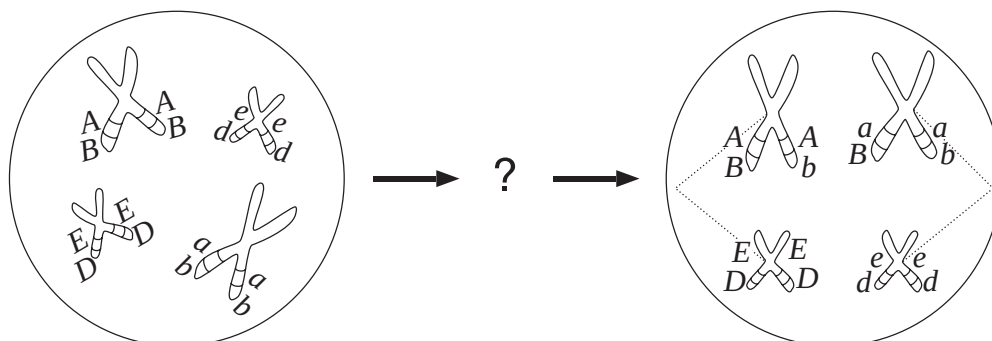
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Question 22 (8 marks)

The diagram shows two steps of the process of meiosis occurring in a cell with four chromosomes.



- (a) Describe the behaviour of the chromosomes between the steps shown. 2

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- (b) List FOUR possible combinations of alleles that would be found in the gametes resulting from this process. 2

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- (c) Explain ONE advantage of the process of meiosis to the species. 2

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- (d) Distinguish between the terms *allele* and *gene*. 2

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Centre Number

Section I – Part B (continued)

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Student Number

Marks**Question 23** (4 marks)

Nothofagus gunnii is a deciduous beech tree that grows in Tasmania. When environmental temperatures decrease at the beginning of winter the beech tree drops its leaves.

4

Design a procedure, using potted beech seedlings, to investigate the temperature at which leaves begin to drop.

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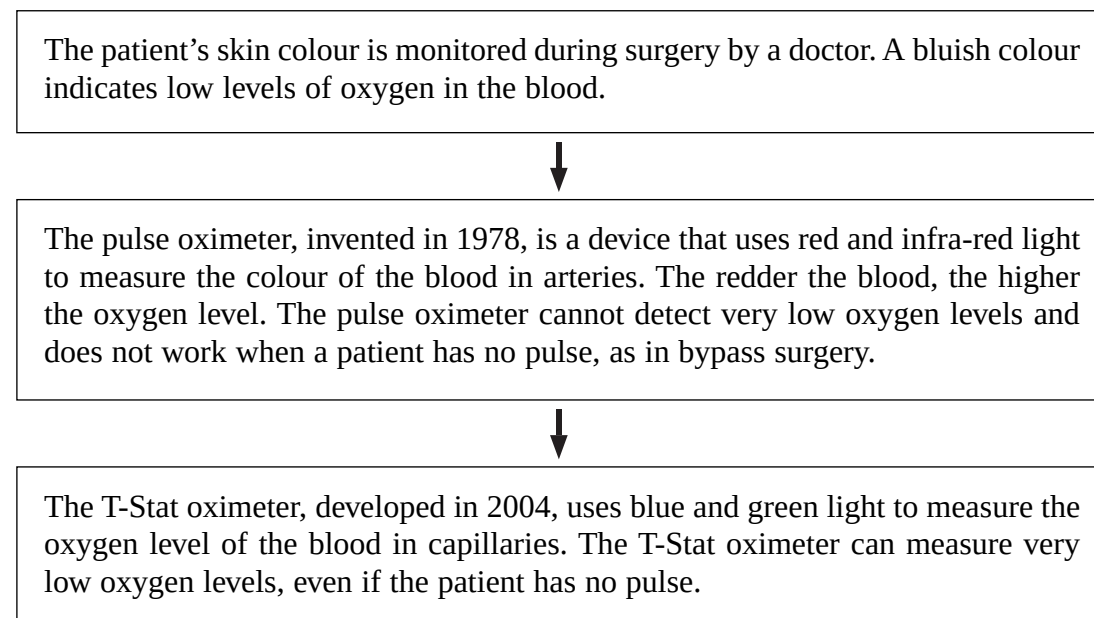
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Question 24 (6 marks)

The flowchart shows the development of technology used to measure oxygen concentration in blood during surgery.



- (a) Why is it important to monitor oxygen levels in the blood during surgery? 1

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- (b) Explain ONE advantage of the T-Stat oximeter over the pulse oximeter. 2

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- (c) Explain TWO changes in the chemical composition of blood as it moves along a capillary. 3

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Centre Number

Section I – Part B (continued)

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Student Number

Marks**Question 25** (3 marks)

Epidemiological studies indicate that there is a relationship between smoking and the incidence of lung cancer.

3

What information would have been gathered to establish this relationship?

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Question 26 (3 marks)

Describe how genes assist in the maintenance of health.

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8

Analyse the possible effects of temperature change on Australian plants and animals with regard to the short-term survival of individuals and long-term survival of species.

[illegible]

Biology

Section II

25 marks

Attempt ONE question from Questions 28–32

Allow about 45 minutes for this section

Answer the question in a writing booklet. Extra writing booklets are available.

	Pages
Question 28 Communication	22–23
Question 29 Biotechnology	24–25
Question 30 Genetics: The Code Broken?	26–27
Question 31 The Human Story	28–29
Question 32 Biochemistry	30–31

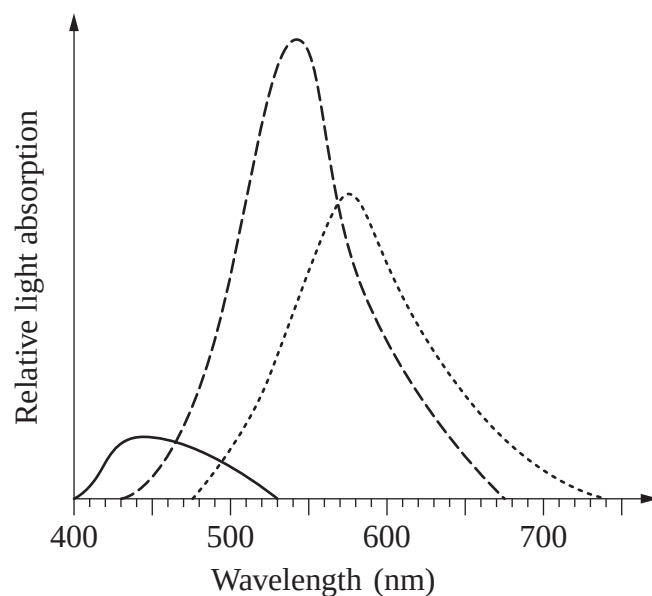
Question 28 — Communication (25 marks)

- | | | | |
|-----|------|--|---|
| (a) | (i) | Name TWO refractive media in the human eye. | 2 |
| | (ii) | Explain how a problem with ONE of these media may contribute to poor eyesight or blindness. | 3 |
| (b) | (i) | Describe an investigation you conducted to distinguish parts of the brain and locate the regions involved in speech, sight and sound perception. | 3 |
| | (ii) | How would you evaluate the relevance and reliability of the information gathered in this investigation? | 3 |
| (c) | | Assess how structures in the human body that produce, detect and perceive sound enable effective communication. Include examples of these structures in your answer. | 7 |

Question 28 continues on page 23

Question 28 (continued)

(d) The graph shows the relative light absorption by cones in the human eye.



KEY

- Blue pigment cone
 ---- Green pigment cone
 Red pigment cone

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 978-0-17-438425-0, first published in 1992.

- | | | |
|-------|---|----------|
| (i) | Estimate the wavelength at which the green pigment cone absorbs most light. | 1 |
| (ii) | Describe the pattern of light absorption by the red pigment cone, and suggest why all three pigment cones are necessary for colour vision. | 2 |
| (iii) | Explain the relationship between the occurrence of colour vision in animals and their use of colour for communication. Include examples in your answer. | 4 |

End of Question 28

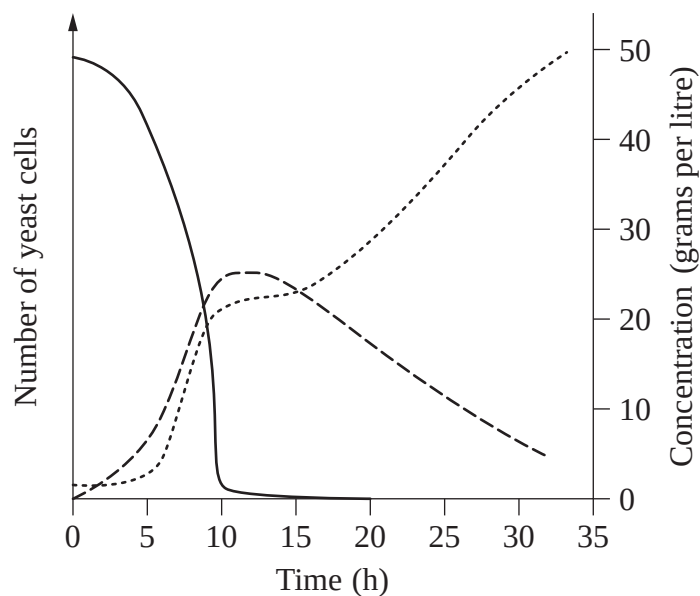
Question 29 — Biotechnology (25 marks)

- | | | | |
|-----|------|--|---|
| (a) | (i) | Name TWO organic compounds produced by fermentation techniques developed since the early eighteenth century. | 2 |
| | (ii) | Explain how the use of ONE of these organic compounds had an impact on society at the time of its introduction. | 3 |
| (b) | (i) | Describe how you gathered and analysed information to outline the purpose of a current application of transgenic technology. | 3 |
| | (ii) | How would you evaluate the relevance and reliability of the information gathered in this investigation? | 3 |
| (c) | | Assess the biotechnological processes used in aquaculture. Include examples in your answer. | 7 |

Question 29 continues on page 25

Question 29 (continued)

(d) The graph shows data recorded during fermentation by yeast cells.



KEY

..... Number of yeast cells

———— Glucose

----- Ethanol

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- (i) Estimate the time at which ethanol concentration is at its maximum. **1**
- (ii) Describe the trend in glucose concentration, and suggest a reason for this change. **2**
- (iii) Explain how changes in technology have modified traditional uses of biotechnology. **4**

End of Question 29

Question 30 — Genetics: The Code Broken? (25 marks)

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|------|---|----------|
| (a) | (i) Name TWO examples of characteristics determined by multiple alleles in an organism other than humans. | 2 |
| | (ii) Explain how ONE of these characteristics provides variability in phenotypes. | 3 |
|
 | | |
| (b) | (i) Describe how you gathered and analysed information to describe the processes used in the cloning of an animal. | 3 |
| | (ii) How would you evaluate the relevance and reliability of the information gathered in this investigation? | 3 |
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 | | |
| (c) | Assess how the mapping of the Human Genome and gene therapy assist in managing a genetic disease OR a form of cancer OR AIDS. | 7 |

Question 30 continues on page 27

Question 30 (continued)

- (d) The graph shows maternal age and incidence of Down syndrome (Trisomy 21) births.



- | | |
|---|----------|
| (i) Estimate the number of infants with Down syndrome per 1000 births for 40-year-old mothers. | 1 |
| (ii) Describe the trend in the number of Down syndrome births, and define the term <i>trisomy</i> . | 2 |
| (iii) Discuss the impact on the genome of transposable genetic elements. | 4 |

End of Question 30

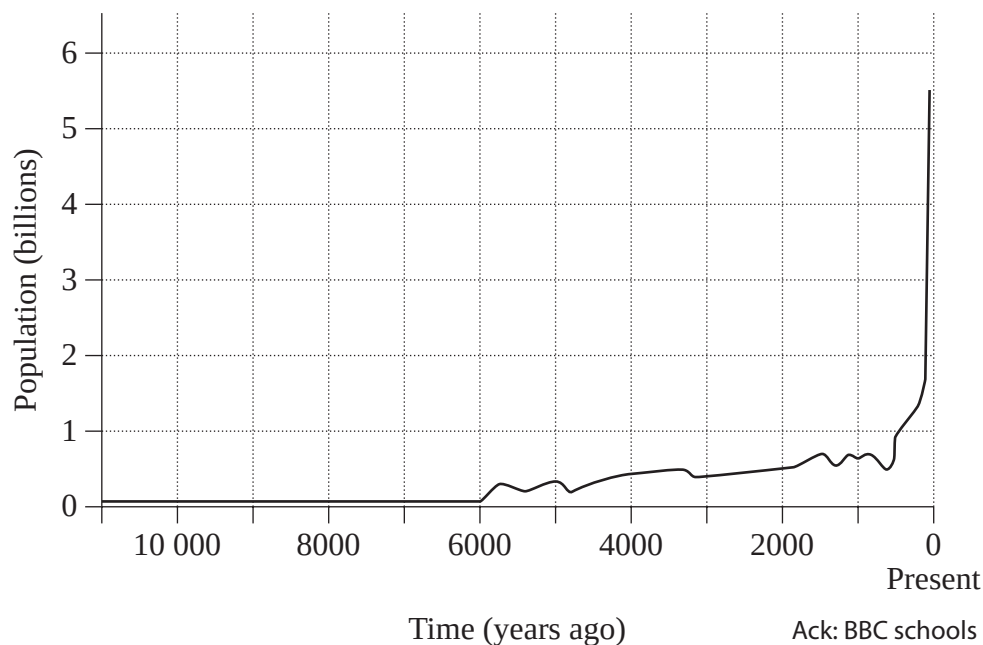
Question 31 — The Human Story (25 marks)

- | | | | |
|-----|------|---|---|
| (a) | (i) | Name TWO hominids known from fossil evidence. | 2 |
| | (ii) | Describe how ONE of these hominids differs from <i>Homo sapiens</i> . | 3 |
| | | | |
| (b) | (i) | Describe how you processed information and used available evidence to assess the contribution of ONE scientist to our increased understanding of human evolution. | 3 |
| | (ii) | How would you evaluate the relevance and reliability of the information gathered in this investigation? | 3 |
| | | | |
| (c) | | Assess the impact of cultural development throughout human evolution. Include examples in your answer. | 7 |

Question 31 continues on page 29

Question 31 (continued)

(d) The graph shows estimated human population for the last 10 000 years.



- (i) Estimate the human population 1000 years ago. 1
- (ii) Describe a change in human population numbers over the last 10 000 years, and suggest ONE reason for this change. 2
- (iii) Discuss the potential impact of a modern technology on future human populations. 4

End of Question 31

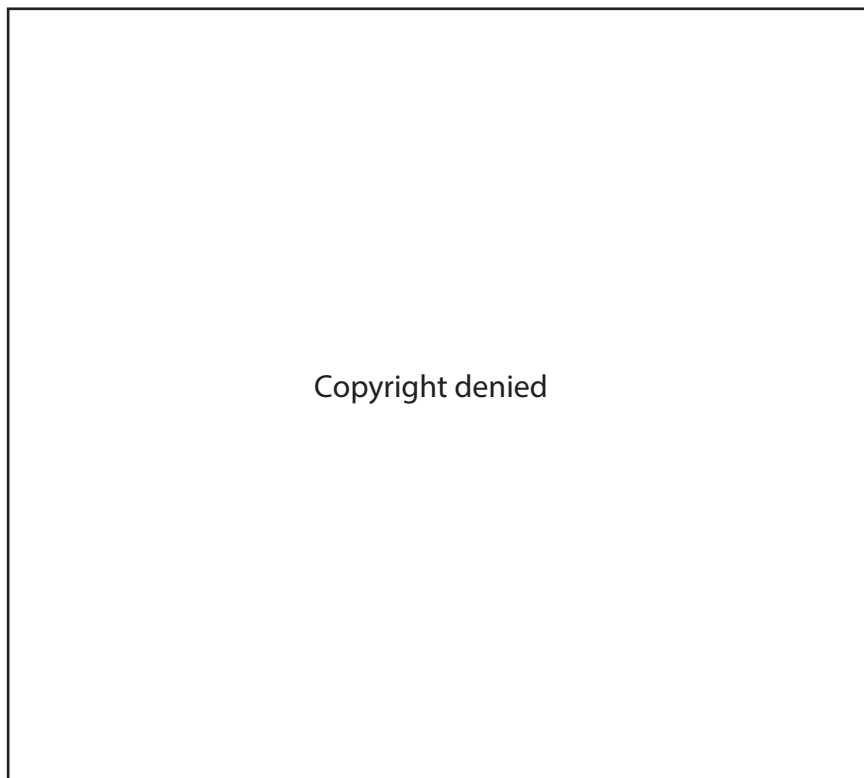
Question 32 — Biochemistry (25 marks)

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|-----|------|---|---|
| (a) | (i) | Name TWO products of photosynthesis. | 2 |
| | (ii) | Explain how ONE of these products is used in living organisms. | 3 |
| (b) | (i) | Describe how you gathered and presented information to compare the size, shape and distribution of chloroplasts in different angiosperms. | 3 |
| | (ii) | How would you evaluate the relevance and reliability of the information gathered in this investigation? | 3 |
| (c) | | Assess the role of isotopes in developing an understanding of photosynthesis. Include examples in your answer. | 7 |

Question 32 continues on page 31

Question 32 (continued)

- (d) The graph shows absorption and action spectra for photosynthesis.



- | | | |
|-------|---|----------|
| (i) | Estimate the wavelength for the maximum rate of photosynthesis. | 1 |
| (ii) | Describe the absorption of chlorophyll <i>a</i> over the spectrum of visible light, and identify the location within a chloroplast where this occurs. | 2 |
| (iii) | Discuss the role of pigments, other than chlorophyll <i>a</i> , in photosynthesis. | 4 |

End of paper

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